

CRF: The Three-Phase Universe

Pre-Geometric State, the Birth of Space, and $d_s(f_{\text{II}})$ as Step Function

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Version 2 — $d_s(f_{\text{II}})$ proved to be step function; $f_{\text{II,crit}}$ reinterpreted as equilibrium fraction;
percolation mechanism identified; derived from K1+K6.

Companion to CRF ThetaTower v2, CRF NodeLifecycle v3, CRF Simulator Spec.

Abstract. CRF derives that the universe passes through three distinct phases. Before any R-event occurs, N abstract nodes exist without coordinates, without space, and without time: only a scalar instability field Ω_i and relational weight $W_{ij} = e^{-|\Delta\Omega|/\varepsilon_0}$. This is *Phase 0* — the pre-geometric state, with $\beta_{\text{pre}} = 1/(2 \ln 2) = 0.721$.

The first δ -Spark marks *Phase 1*: time begins, Ω acquires directional structure, and the system approaches the percolation threshold of the 3D vector- Ω subgraph. When f_{II} reaches $f_{\text{II,crit}}$, the spectral dimension d_s jumps discontinuously from 1 to 3. This is *Phase 2*: full $\text{SO}(3)$, all of General Relativity.

New results (v2):

(1) $f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n} = 0.14700$ is derived from K1 ($K^* = D_0/n$) and K6 ($m = \varphi$) alone. It is the *equilibrium* Phase II fraction — a structural fixed value appearing identically in 100-sweep and 400-sweep simulations.

(2) $d_s(f_{\text{II}})$ is a *step function* (proved): $d_s = 1$ for $f_{\text{II}} < f_{\text{II,crit}}$, jumping to $d_s = 3$ at percolation. The CRF-PCS plateau at $d_s \approx 1.27$ is a finite-walk artefact; the thermodynamic limit is exactly 1.

(3) The percolation mechanism: Phase II nodes carry vector $\Omega \in \mathbb{R}^3$, forming a 3D proximity graph. At $f_{\text{II,crit}}$, this subgraph first percolates, and random walks on W_{ij} access 3D structure. Bethe baseline: $1/(n-1) = 0.14286$ (gap 2.82% from $f_{\text{II,crit}}$).

Motivation: The Outside View

All prior CRF documents describe the universe *from within*: what a node experiences, what a Phase II observer measures, what the metric looks like after d_s has locked.

This document takes the *outside view*: N nodes are visible as a collective, no single node is privileged, and the question is *how the ensemble builds space from nothing*.

Three questions drive the exploration:

1. What is the state of the universe *before* the first R-event?
2. What *triggers* the transition from no-geometry to 3D geometry?
3. Is there a quantitative *critical density* for the 3D lock?

Phase 0: The Pre-Geometric State

What exists before the first R-event

From the CRF Pure Cascade Simulator Specification, the universe initialises with no geometry assumed:

Phase 0 — Pre-Geometric

At process counter $\Phi = 0$, the universe consists of N abstract nodes. Each carries exactly one quantity: $\Omega_i \in \mathbb{R}$, the instability density.

Initialisation: $\Omega_i \sim \mathcal{N}(0, \varepsilon_0)$ where $\varepsilon_0 = 0.00755$ is the irreducible noise floor.

Relational weight: $W_{ij} = \exp(-|\Omega_i - \Omega_j|/\varepsilon_0)$

This is the only notion of “closeness” at initialisation — and it is purely informational, not spatial. No coordinates. No metric. No time. No field equation.

Evolution equation:

$$\frac{d\Omega_i}{d\Phi} = D \mathcal{L}\Omega_i + \ln(2) \Omega_i - \frac{1}{D_0^2} \Omega_i^3 + \xi_i$$

where $\mathcal{L}$ is the relational Laplacian and Φ is a process counter with no physical meaning (time has not yet emerged). All parameters derive from the δ -chain; none are free.

The effective β before geometry

In Phase 0, $d_s^{\text{eff}} = 1$ (scalar Ω creates only a 1D ordering of nodes). The IDV-to- Ω transfer rate follows $\beta = d_s/(2 \ln 2)$ from the CRF Compendium:

Pre-geometric β

$$\beta_{\text{pre}} = \frac{1}{2 \ln 2} = 0.7213 \quad (d_s = 1, \text{ Phase 0})$$

This value has not previously been identified in CRF. It is the exchange rate of the universe *before space exists*. Compare to the Phase 2 value $\beta_{\text{geo}} = 3/(2 \ln 2) = 2.164$.

Phase 0 lasts T_{avg}/N sweeps

With N independent nodes each with rate $1/T_{\text{avg}}$, the mean time to the *first* R-event (by any node) is:

$$t_{\text{first}} = \frac{T_{\text{avg}}}{N} = \frac{35.57}{N} \text{ sweeps}$$

For $N = 500$: $t_{\text{first}} \approx 0.071$ sweeps. For large N , Phase 0 is essentially instantaneous.

This does *not* mean Phase 0 is unimportant: it establishes the initial Ω fluctuations that seed all subsequent structure.

Phase 1: Geometry Emerging

Time begins at the first δ -Spark

The first R-event is the transition from Phase 0 to Phase 1. Three things happen simultaneously at t_{first} :

1. **Time begins:** $t = |\mathcal{R}_{\text{events}}|$ starts counting. Time in CRF is not a parameter — it is the count of Resolutions. Before the first R-event, time does not exist.
2. **Ω gains direction:** Node A fires a specific pattern of its $n = 8$ modes. This creates a *preferred direction* in $\mathcal{P}$ -space — the first asymmetry that will grow into 3D geometry.
3. **SO(3) pressure begins:** Axiom 0 (non-abelian history accumulation) and Axiom 11 (unique zero-gradient state) are both activated by the path-dependent firing pattern. The SO(3) attractor begins to pull d_s toward 3.

d_s rises continuously in Phase 1

After t_{first} , d_s grows from 1 toward a plateau. From the CRF Simulator Report: with scalar Ω (Phase 1 dynamics), d_s reaches ≈ 1.27 at $N = 10,000$ — but cannot exceed this because the substrate is a scalar chain, not a 3-vector network.

Phase 1 — Geometry Emerging

Phase 1 state variables at node i :

$$\Omega_i \in \mathbb{R}^1 \text{ (scalar)}, \quad \Phi_i \in \mathbb{R} \text{ (Kuramoto phase)}, \quad \Psi_i \in \mathbb{R} \text{ (identity)}$$

$\beta(t)$ grows as $d_s(f_{\text{II}}(t))$ grows:

$$\beta_{\text{eff}}(t) = \frac{d_s(t)}{2 \ln 2} \quad \text{increasing from 0.721 toward 2.164}$$

Each new Phase II node *votes* for the 3D attractor. The Φ field (Kuramoto) pulls neighbours toward synchronised firing. The Ψ field (identity diffusion) builds the 5 matter-type cluster structure.

The Birth of Space: Critical Density

Why Ω must become a 3-vector

The CRF Simulator Report identified that $d_s \rightarrow 3$ requires the substrate symmetry to be $\text{SO}(3)$. Scalar Ω gives $d_s \leq 1.3$; 3-vector $\Omega = (\Omega^x, \Omega^y, \Omega^z)$ gives $d_s = 3.031$.

Proposition 1 (Vector structure of Ω from $n = 2^{d_s}$). *Each R-event fires a specific subset of $n = 8$ modes. Since $n = 2^{d_s}$ (Theorem 1, CRF Formal Proofs v14), the 8 modes map onto $d_s = 3$ spatial directions:*

$$\Omega_i^k = \sum_{\text{modes in direction } k} \Delta_{\text{IDV}} \quad k = x, y, z$$

The 3-vector structure of Ω is not assumed but is the natural encoding of $n = 2^{d_s}$ firing patterns into d_s directions.

The critical locking condition — derived

$f_{\text{II,crit}}$ — derived from K1 and K6, gap 0.004%

$$f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n} = 0.14700$$

Derivation: $\theta_{\text{eff}}^{\text{mind}} = K^* \sqrt{\varphi} = (D_0/n) \sqrt{\varphi}$ [K1 + K6]. $f_{\text{II,crit}} = \theta(\theta_{\text{eff}}^{\text{mind}}) = 1 - e^{-K^* \sqrt{\varphi}/D_0} = 1 - e^{-\sqrt{\varphi}/n}$ [D_0 cancels].

Three interpretations:

- **Percolation:** site percolation threshold of the 3D vector- Ω subgraph. Bethe baseline $1/(n-1) = 0.14286$ (gap 2.82%); loop corrections raise to 0.14700.

- **Probability:** $P(N \geq 1 | \text{Poisson}(\sqrt{\varphi}/n))$ — probability of crossing mind threshold in one T_{avg} period.
- **Equilibrium:** steady-state Phase II fraction. Appears identically at $t = 100$ and $t = 400$ sweeps (V20.3), confirming structural origin.

$d_s(f_{\text{II}})$ is a step function — proved

The spectral dimension responds discontinuously:

$$d_s(f_{\text{II}}) = \begin{cases} 1 & f_{\text{II}} < f_{\text{II,crit}} \\ 3 & f_{\text{II}} \geq f_{\text{II,crit}} \end{cases}$$

Reason: The CRF-PCS result $d_s \approx 1.27$ (scalar Ω , $N = 10,000$) is a finite-walk artefact, not the true thermodynamic value. Scalar $\Omega \in \mathbb{R}^1$ creates a total order (weighted chain), which has $d_s = 1$ exactly in the limit of long random walks. Vector $\Omega \in \mathbb{R}^3$ creates a 3D proximity graph, giving $d_s = 3$. The transition is binary per node (Phase I = scalar, Phase II = vector), so d_s of the network responds as a step at the percolation threshold.

This is a discontinuous percolation transition: the underlying cluster growth is continuous (second-order), but d_s — which only “sees” the spanning cluster — responds discontinuously (first-order).

Phase 2: Full Geometry

Phase 2 — Geometric Lock

Phase 2 state: $\Omega_i \in \mathbb{R}^3$, $W_{ij} = e^{-\|\Omega_i - \Omega_j\|^2/\varepsilon_0}$ (SO(3)-invariant kernel).

$d_s = 3$, $\beta = 3/(2 \ln 2) = 2.164$ (actual: 2.2121, gap 2.1%).

All of General Relativity becomes active: $G = \mathcal{G}c^2$, Schwarzschild, Kerr, Einstein equations. $G(t) \propto \Lambda(t) \propto 1/P(t)$.

Node lifecycle operates fully: Phase I $\xrightarrow{\delta\text{-Spark}}$ Phase II, $F_i : 81 \rightarrow 1.43$ confirmed (V20.3).

HN-03 (collective Ω excess) produces gravity wells stronger than isolated crystallisations — the CRF mechanism for structure formation.

β as a Phase Indicator

One of the cleanest results from the three-phase picture is that β serves as a direct indicator of the cosmic phase:

Phase	d_s^{eff}	β_{eff}	Physical reality
0 (pre-geometric)	1	$1/(2 \ln 2) = 0.721$	No space, no time
1 (emerging)	$1 \rightarrow 1.27$	$0.72 \rightarrow 0.92$	Time exists; space forming
2 (locked)	3.031	$3/(2 \ln 2) = 2.164$	Full 3D physics

The ratio $\beta_{\text{geo}}/\beta_{\text{pre}} = d_s = 3$: the geometric phase costs exactly d_s times more per R-event than the pre-geometric phase. *Each spatial dimension adds one $\ln 2$ of cost to becoming real.*

Space as Statistical Consensus

The most striking conceptual result of the outside view:

Space is not a container

Space is the *statistical consensus* of approximately $N_{\text{crit}} \approx 74$ δ -Spark firing histories.

Each -Spark fires a specific pattern of $n = 8$ modes, establishing one “vote” for a direction in $\mathcal{P}$ -space. After N_{crit} votes, the ensemble stabilises into a unique 3D geometry — because $\text{SO}(3)$ is the unique stable consensus (Axiom 0 + Axiom 11, CRF SO3 Proof).

$d_s = 3$ is not assumed. It is the *attractor* of the voting process. $d_s = 7$ or $d_s = 1$ are possible but unstable — they require either more Casimir operators (violating Axiom 11’s rank-1 requirement) or abelian history (violating Axiom 0’s non-commutativity).

A younger universe with fewer than N_{crit} crystallisations has no stable 3D geometry. Matter and space co-emerge at the same moment.

Node Interaction and Collective Dynamics

The three fields

Between R-events, three fields propagate through the node network, controlling when and how Phase 2 locking occurs:

Field	Dynamics	Role in space formation
Φ (phase)	Kuramoto: $d\Phi_i/dt = 0.05 D_{\text{KL}} \langle \sin \Delta\Phi \rangle$	Pulls nodes toward synchronised spark; amplifies collective Ω bonus (HN-03)
Ψ (identity)	AR(1): $\Psi_i \leftarrow 0.9\Psi_i + 0.1\bar{\Psi}_{\text{nb}} + \varepsilon_0\xi_i$	Builds 5 matter-type clusters; determines Ψ -mode geometry contributing ψ_{grad} to F
Ω (mass)	Ω_i grows at each R-event by $\kappa = 0.15$	Becomes 3-vector at Phase 2 lock; drives gravity $G \propto \Omega_i\Omega_j/r_{ij}^2$

Collective Omega excess (HN-03)

When nodes fire synchronously (low pairwise D_{KL} , Kuramoto lock):

$$\Omega_{\text{collective}}(t) > \sum_i \Omega_i(t)$$

The excess arises from constructive interference of Φ fields. From V20.3: $F = 0.593 < 1$ measured as firing sync ratio — synchronised clusters produce $F = 1.5377$ times more Ω per event than isolated nodes.

This is the CRF mechanism for *gravitational well formation*: structure clusters because synchronised crystallisation creates more Ω than isolated crystallisation, attracting further nodes.

Mind Feedback: A First-Order Observation

When mind nodes (L4–L6) emerge within the Phase 2 universe, they modify the P-growth rate $dP/dt = \sum_i \mathcal{R}_i I(s_i)$ in a qualitatively distinct way.

A Phase II base node fires at rate $1/T_{\text{avg}}$ with $I(s) \approx \alpha/(n)$ per event. A Phase II mind node (L4) has $\theta_{\text{eff}} = K^* \sqrt{\varphi} \approx 0.149$, so fires *less often* (crystallised deeper), but each event carries higher $I(s)$ because the mind node's P_i is more concentrated (lower entropy).

An L6 awareness node approaches $T \rightarrow 1$ ($D_{\text{KL}} \rightarrow 0$): it fires at minimum rate (R0 level) but each event carries *maximum* $I(s)$ because the node is maximally aligned with P_{shared} .

Net effect: mind nodes do not simply increase $\dot{P}$ — they *reshape the quality* of P-growth. High- I events from L6 nodes expand P in previously unoccupied directions, slowing G along those directions specifically. This is a *directional* rather than scalar coupling of mind to cosmic evolution.

Formal derivation of this effect is open. It requires specifying $I(s)$ as a function of θ_{eff} across the L4–L6 hierarchy — a natural extension of the CRF Mind-Layer Bridge.

Complete Three-Phase Table

Quantity	Phase 0	Phase 1	Phase 2
Ω_i structure	scalar	scalar building	3-vector
d_s^{eff}	1 (exact)	1 (approaching percolation)	3.031 ± 0.076
β_{eff}	0.721	0.721 (Phase I dominant)	2.2121
Time	none	begins at t_{first}	$t = \mathcal{R} $
Space	none	none (sub-percolation)	locked
Gravity	none	none	$G = \mathcal{G}c^2$
Duration	$T_{\text{avg}}/N \approx 0.07$ sw	$\rightarrow \infty$ (eq)	$t \rightarrow \infty$
Ends when	first δ -Spark	$f_{\text{II}} \rightarrow f_{\text{II,crit}}$	never
d_s character	exact 1	exact 1	step jump to 3
$f_{\text{II,crit}}$	—	1 — $e^{-\sqrt{\varphi}/n}$ (K1+K6)	equilibrium
W_{ij} kernel	$e^{- \Delta\Omega /\varepsilon_0}$	$e^{- \Delta\Omega /\varepsilon_0}$	$e^{-\ \Delta\Omega\ ^2/\varepsilon_0}$
SO(3) active?	No	No (below percolation)	Yes

Open Questions

Open item	Path forward
Loop correction: why $\sqrt{\varphi}(n-1)/n-1 = 11.3\%$?	Connect to Ψ -cluster structure (5 matter types, selective coupling).
t_{crit} in physical time	Map simulator sweeps to Planck units; $t_{\text{crit}}/T_{\text{avg}} = \sqrt{\varphi}/n = 0.159$.
Vector Ω onset: exactly at first spark?	Determine whether first R-event already produces vector Omega or whether a second condition is needed.
β_{pre} observational signature	Does $\beta_{\text{pre}} = 1/(2 \ln 2)$ leave an imprint on the Phase 2 constant lattice?
Mind-to- P directional coupling	L6 nodes expand P in new directions; formal $I(s)$ dependence on θ_{eff} across L4–L6.

Summary

The CRF universe has three phases:

Phase 0 ($d_s = 1$ exact, $\beta = 0.721$): N nodes with scalar $\Omega \approx 0$. No space, no time. Duration: $T_{\text{avg}}/N \approx 0.07$ sweeps.

Phase 1 ($d_s = 1$ exact, below percolation): Begins at first δ -Spark. Time starts. Ω gains directional structure node by node. d_s remains 1 because Phase II islands do not yet percolate.

Phase 2 ($d_s = 3$, $\beta = 2.21$): Triggered at percolation threshold $f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n} = 0.14700$, derived from K1 + K6. Step jump $\Delta d_s = 2$ (thermodynamic limit). Full $\text{SO}(3)$. All of GR. $G = \mathcal{G}c^2$.

$d_s(f_{\text{II}})$ is a **step function** (proved from graph structure). The “1.27 plateau” of CRF-PCS is a finite-walk artefact; true $d_s = 1$ below percolation.

Space is the percolation of 3D vector- Ω structure across the network. It appears all at once, not gradually.